

Assignment 2

Encoder - $W \in \mathbb{R}^{k \times d}$

Decoder - $V \in \mathbb{R}^{d \times k}$

$$L = \sum_{n=1}^N \|x^{(n)} - \hat{x}^{(n)}\|^2$$

$$\hat{x}^{(n)} = V W x^{(n)}$$

$$V = W^T$$

$$L = \sum_{n=1}^N \|x^{(n)} - V W x^{(n)}\|^2$$

$$= \sum_{n=1}^N \|x^{(n)} - W^T W x^{(n)}\|^2$$

$$= \sum_{n=1}^N \|x^{(n)}\|^2 - \text{trace} \left(W^T \left(\sum_{n=1}^N x^{(n)} x^{(n)T} \right) W \right)$$

$$\arg \min_W L = \arg \max_W \text{trace} \left(W^T \left(\sum_{n=1}^N x^{(n)} x^{(n)T} \right) W \right)$$

$$\text{Let } \frac{1}{N} \sum_{n=1}^N x^{(n)} x^{(n)T} = X X^T$$

Using Singular Value Decomposition :

$$X = U \Sigma V$$

- Where -
- U contains left eigenvectors of X (unitary)
 - Σ is a diagonal matrix with eigenvalues
 - V contains right eigenvectors of X (unitary)

$$X X^T = U \Sigma V V^T \Sigma^T U^T$$

$$W^T X X^T W = W^T U \Sigma V V^T \Sigma^T U^T W$$

$$= W^T U \Sigma \Sigma^T U^T W$$

Setting W to contain the first k eigenvectors corresponding to the largest k eigenvalues and U to also contain the first k eigenvectors corresponding to the largest k eigenvalues

$$\Rightarrow W^T U = I$$

Σ contains the top k eigenvalues in the diagonal

$$\Rightarrow \cancel{\arg \min_W L = \arg \max_W \text{trace}(\Sigma \Sigma^T)} = \arg \max$$

$$\text{trace}(\Sigma \Sigma^T) = \text{sum of squares of top } k \text{ eigenvalues}$$

$$\geq \text{sum of squares of any } k \text{ eigenvalues}$$

$\therefore W^*$ forms an orthonormal basis for the top k principal components of the data covariance matrix

$$\Sigma = \frac{1}{N} \sum_{n=1}^N x^{(n)} x^{(n)T} = X X^T$$